

## Chapter 9 - The Voodoo 3D Rasteriser

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Voodoo is Intuition Engine's hardware triangle rasteriser. It draws flat-shaded, Gouraud-shaded, textured, depth-tested, alpha-tested, fogged, chroma-keyed, and dithered triangles into an RGBA framebuffer with an associated Z buffer. Voodoo is compositor layer 20, so its visible pixels sit above the other picture sources.

The programmer's path is the same as the rest of this guide: type BASIC commands first, then use POKE, PEEK, POKE8, and PEEK8 when direct register control is useful. The register names keep their Voodoo heritage, but the addresses, formats, side effects, and examples below are the Intuition Engine contract.

### 9.1 What Voodoo can show

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Use Voodoo when a picture is easier to describe as triangles than as tiles or rectangles: a spinning sign, a shaded cockpit panel, a textured floor, a depth-tested overlay, or a sprite-like quad that needs alpha. It is still one card on the same bus. BASIC sets up the state, vertices become fixed-point register values, and the compositor places the finished frame above the other display chips.

| Item               | Value                                |
|--------------------|--------------------------------------|
| Output             | RGBA framebuffer with depth buffer   |
| Default resolution | 640 by 480                           |
| Maximum resolution | 800 by 600                           |
| Primitive          | Triangle, defined by three vertices  |
| Per-vertex data    | Position, RGBA, Z, S, T, and W       |
| Texture memory     | 64 KB at \$D0000                     |
| Texture sizes      | up to 256x256 in the texture memory  |
| Texture formats    | paletted, intensity, alpha, and ARGB |
| Depth functions    | 8 functions, numbered 0 to 7         |
| Alpha test/blend   | Yes                                  |
| Fog                | Per-pixel depth fog                  |
| Chroma key         | Yes                                  |
| Dither             | Ordered 4x4 or 2x2                   |

### 9.2 Programming model

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Voodoo is state-machine driven:

1. Enable the chip with VOOD00 ON.
2. Set the picture size with VOOD00 DIM.
3. Set draw state, usually FBZ\_MODE, ALPHA\_MODE, and FBZCOLOR\_PATH.
4. Clear the back buffer with VOOD00 CLEAR.

5. Write vertices with VERTEX A, VERTEX B, and VERTEX C.
6. Write colour, depth, texture, or W values with V00D00 TRI . . . commands.
7. Submit the triangle with TRIANGLE.
8. Publish the frame with V00D00 SWAP.

The submission point matters. When TRIANGLE writes TRIANGLE\_CMD, Voodoo latches the current raster state for that triangle: draw mode, alpha mode, colour combine, fog, chroma key, stipple, clip rectangle, slopes, texture mode, and the currently uploaded texture. Later writes to those registers, or a later TEXTURE UPLOAD, affect later triangles only. This lets one batch mix plain, shaded, textured, fogged, and clipped triangles before one final V00D00 SWAP.

The simplest triangle is a typed BASIC program:

```
10 REM V00D00 FIRST TRIANGLE
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008200
50 V00D00 CLEAR &HFF101020
60 VERTEX A 320,70
70 VERTEX B 540,390
80 VERTEX C 100,390
90 V00D00 TRICOLOR 4096,0,0,4096
100 TRIANGLE
110 V00D00 SWAP
```

Line 40 enables RGB writes and selects the back draw target. Line 90 uses 12.12 fixed-point colour: 4096 means full intensity and 0 means off. The result is a red triangle on a dark background.

## 9.3 BASIC keywords

The V00D00 keyword introduces most Voodoo subcommands. VERTEX, TRIANGLE, ZBUFFER, and TEXTURE are companion hardware verbs that operate on the same state.

| Form                           | Effect                                                                                         |
|--------------------------------|------------------------------------------------------------------------------------------------|
| V00D00 ON / V00D00 OFF         | Enable or disable the chip.                                                                    |
| V00D00 DIM w,h                 | Set output dimensions and resize the buffers if valid.                                         |
| V00D00 CLEAR colour            | Store a 32-bit clear word in COLOR0 and trigger FAST_FILL_CMD.                                 |
| V00D00 SWAP                    | Flush queued triangles and publish the frame.                                                  |
| V00D00 CLIP x0,y0,x1,y1        | Set the scissor rectangle.                                                                     |
| V00D00 COMBINE mode            | Write FBZCOLOR_PATH. Common values are 0 iterated, 1 texture, 97 modulate.                     |
| V00D00 LFB mode                | Write LFB_MODE.                                                                                |
| V00D00 PIXEL x,y,word          | Write one 32-bit word into the linear texture/LFB aperture at \$D0000 + ((y * width + x) * 4). |
| V00D00 RGB ON / V00D00 RGB OFF | Set or clear the RGB-write bit in FBZ_MODE.                                                    |
| V00D00 TRICOLOR r,g,b[,a]      | Write START_R, START_G, START_B, and optional START_A. Values are 12.12 fixed point.           |

| Form                                                  | Effect                                                                               |
|-------------------------------------------------------|--------------------------------------------------------------------------------------|
| V00D00 TRISHADE<br>drdx,drdy,dgdx,dgdy,dbdx,dbdy      | Write colour gradient registers.                                                     |
| V00D00 TRIDEPTH start_z,dzdx,dzdy                     | Write depth start and gradient registers. Values are 20.12 fixed point.              |
| V00D00 TRIUV<br>start_s,start_t,dsdx,dt dx,dsdy,dt dy | Write texture coordinate start and gradient registers. Values are 14.18 fixed point. |
| V00D00 TRIW start_w,dwdx,dwdy                         | Write perspective W start and gradient registers. Values are 2.30 fixed point.       |
| V00D00 ALPHA TEST ON / OFF                            | Set or clear alpha-test enable.                                                      |
| V00D00 ALPHA FUNC n                                   | Set alpha-test function bits 1 to 3.                                                 |
| V00D00 ALPHA BLEND ON / OFF                           | Set or clear alpha-blend enable.                                                     |
| V00D00 ALPHA SRC n / V00D00 ALPHA DST n               | Set source and destination blend factors.                                            |
| V00D00 FOG ON / OFF                                   | Set or clear fog enable.                                                             |
| V00D00 FOG COLOR r,g,b                                | Set fog colour as an RGB triplet.                                                    |
| V00D00 DITHER ON / OFF                                | Set or clear ordered dither.                                                         |
| V00D00 CHROMAKEY ON / OFF                             | Set or clear chroma-key discard.                                                     |
| V00D00 CHROMAKEY COLOR r,g,b                          | Set the chroma-key colour.                                                           |
| VERTEX A x,y                                          | Write vertex A, converting integer coordinates to 12.4 format.                       |
| VERTEX B x,y                                          | Write vertex B.                                                                      |
| VERTEX C x,y                                          | Write vertex C.                                                                      |
| TRIANGLE                                              | Queue the current triangle.                                                          |
| ZBUFFER ON / OFF                                      | Set or clear depth-test enable.                                                      |
| ZBUFFER FUNC n                                        | Set depth function bits 5 to 7.                                                      |
| ZBUFFER WRITE ON / OFF                                | Set or clear depth-buffer writes.                                                    |
| TEXTURE ON / OFF                                      | Set or clear texture enable in TEXTURE_MODE.                                         |
| TEXTURE MODE mode                                     | Write TEXTURE_MODE.                                                                  |
| TEXTURE BASE lod,addr                                 | Write one texture base register.                                                     |
| TEXTURE DIM w,h                                       | Set upload width and height.                                                         |
| TEXTURE UPLOAD                                        | Commit texture memory into the texture sampler.                                      |

## 9.4 Data formats

Voodoo registers are 32-bit words. POKE32 writes a whole word. PEEK32 reads a whole word. POKE8 may be used for byte work, but register side effects happen only after byte 3 of a word has been written.

| Value      | Format | Unit          |
|------------|--------|---------------|
| Vertex X/Y | 12.4   | 1 pixel is 16 |

| Value                  | Format | Unit                   |
|------------------------|--------|------------------------|
| R/G/B/A                | 12.12  | full intensity is 4096 |
| Z                      | 20.12  | 1.0 is 4096            |
| S/T texture coordinate | 14.18  | 1.0 is 262144          |
| W                      | 2.30   | 1.0 is 1073741824      |

VERTEX A, VERTEX B, and VERTEX C shift integer coordinates left by four for you. V00D00 TRICOLOR, V00D00 TRIDEPTH, V00D00 TRIUV, and V00D00 TRIW write the raw fixed-point words you supply.

V00D00 CLEAR and COLOR0 use a 32-bit ARGB word: \$AARRGGBB. The texture memory aperture is byte-addressed RGBA. When a 32-bit POKE32 writes one RGBA texel into little-endian texture memory, the word appears as \$AABBGRR. For example, &HFF0000FF stores an opaque red texture pixel.

## 9.5 Register block

Voodoo's register block runs from \$F8000 to \$F87FF. Registers are 32-bit words at 4-byte aligned addresses.

### 9.5.1 Control and status

| Address | Name          | Purpose                 |
|---------|---------------|-------------------------|
| \$F8000 | V00D00_STATUS | Status.                 |
| \$F8004 | V00D00_ENABLE | Bit 0 enables the chip. |

V00D00\_STATUS bits:

| Bits  | Field    | Meaning                              |
|-------|----------|--------------------------------------|
| 0     | FBI_BUSY | Framebuffer interface busy.          |
| 1     | TMU_BUSY | Texture unit busy.                   |
| 2     | SST_BUSY | Overall chip busy.                   |
| 6     | VRETRACE | Vertical retrace active.             |
| 7     | SWAPBUF  | Buffer swap pending.                 |
| 12-19 | MEMFIFO  | Coarse memory FIFO free-space field. |
| 20-24 | PCIFIFO  | Command FIFO free-space field.       |

MEMFIFO is useful as a ready/not-ready field, not as a cycle-accurate historical FIFO depth. It is non-zero while the queued triangle batch can accept more triangles. It reads zero only while the current batch is exactly at its 4096 triangle limit. The next TRIANGLE\_CMD at that point flushes the full batch into the drawing buffer without publishing the frame, then starts a fresh batch for the new triangle. PCIFIFO reports command space for the current high-level engine path.

FBI\_BUSY and SST\_BUSY are set while Voodoo is rendering a queued batch or publishing a frame. SWAPBUF is set while a publish swap is pending. Up to two swap jobs may be in flight, so a later SWAP\_BUFFER\_CMD waits only when that small pipeline is full. A simple BASIC program may not see these bits because small frames finish before the next PEEK, but machine code and IE Mon can still poll them around a large frame. TRIANGLE\_CMD itself normally only appends work to the batch; when the batch is already full it first flushes that full batch as a render-only step.

## 9.5.2 Vertex and attribute registers

| Address            | Name                   | Format            |
|--------------------|------------------------|-------------------|
| \$F8008 to \$F801C | VERTEX_AX to VERTEX_CY | 12.4 fixed point  |
| \$F8020            | START_R                | 12.12 fixed point |
| \$F8024            | START_G                | 12.12 fixed point |
| \$F8028            | START_B                | 12.12 fixed point |
| \$F802C            | START_Z                | 20.12 fixed point |
| \$F8030            | START_A                | 12.12 fixed point |
| \$F8034            | START_S                | 14.18 fixed point |
| \$F8038            | START_T                | 14.18 fixed point |
| \$F803C            | START_W                | 2.30 fixed point  |
| \$F8040 to \$F805C | DRDX to DWDX           | per-X gradients   |
| \$F8060 to \$F807C | DRDY to DWDY           | per-Y gradients   |

Without `COLOR_SELECT`, every submitted triangle uses the current start values for all three vertices. Writing `COLOR_SELECT` with 0, 1, or 2 selects vertex A, B, or C as the target for later `START_*` writes. The selection applies to colour, Z, S, T, and W. After `TRIANGLE`, Gouraud selection is reset for the next triangle.

## 9.5.3 Command registers

| Address | Name            | Purpose                                                                                  |
|---------|-----------------|------------------------------------------------------------------------------------------|
| \$F8080 | TRIANGLE_CMD    | Queue one triangle using current state.                                                  |
| \$F8084 | FTRIANGLECMD    | Queue one triangle through the fast-triangle alias.                                      |
| \$F8088 | COLOR_SELECT    | Select vertex 0, 1, or 2 for later attribute writes.                                     |
| \$F8120 | NOP_CMD         | No operation.                                                                            |
| \$F8124 | FAST_FILL_CMD   | Clear the drawing buffer with <code>COLOR0</code> .                                      |
| \$F8128 | SWAP_BUFFER_CMD | Flush triangles and publish the frame. Bit 0 waits for retrace; bit 1 clears after swap. |

`TRIANGLE_CMD` queues work. It does not rasterise visible pixels and it does not wait for the rasteriser to draw the triangle. The current vertices, per-vertex attributes, raster state, and currently uploaded texture are latched into the triangle batch, up to 4096 triangles. Later writes to `FBZ_MODE`, `ALPHA_MODE`, `FBZCOLOR_PATH`, `TEXTURE_MODE`, fog, chroma, stipple, clip, slope, or texture upload state do not change triangles that are already queued.

If the batch is already full, the next `TRIANGLE_CMD` renders that full batch into the drawing buffer without publishing the frame, clears the batch, and then queues the new triangle. This is how very large frames remain complete.

Pixels appear after `SWAP_BUFFER_CMD`. That command hands the queued triangles to the rasteriser using the state each one latched when it was submitted, clears the batch, swaps buffers, and publishes the frame to the compositor. It returns while render or publish work may still be active. Voodoo can keep two swap jobs in flight, so a later swap waits only when that pipeline is full. During render or publish work the status register reports `FBI_BUSY` and `SST_BUSY`; during a pending publish it also reports `SWAPBUF`.

## 9.5.4 Mode and state

| Address | Name            | Purpose                                    |
|---------|-----------------|--------------------------------------------|
| \$F8104 | FBZCOLOR_PATH   | Colour combine and source selects.         |
| \$F8108 | FOG_MODE        | Fog pipeline.                              |
| \$F810C | ALPHA_MODE      | Alpha test and alpha blend.                |
| \$F8110 | FBZ_MODE        | Framebuffer, Z, clipping, and draw enable. |
| \$F8114 | LFB_MODE        | Linear framebuffer mode latch.             |
| \$F8118 | CLIP_LEFT_RIGHT | (left << 16)   right.                      |
| \$F811C | CLIP_LOW_Y_HIGH | (top << 16)   bottom.                      |

FBZ\_MODE bits:

| Bit   | Field        | Meaning                                              |
|-------|--------------|------------------------------------------------------|
| 0     | CLIPPING     | Enable scissor.                                      |
| 1     | CHROMAKEY    | Enable chroma-key discard.                           |
| 2     | STIPPLE      | Enable stipple mask.                                 |
| 3     | WBUFFER      | Use W-buffer interpretation for depth normalisation. |
| 4     | DEPTH_ENABLE | Enable depth test.                                   |
| 5-7   | DEPTH_FUNC   | Depth compare function.                              |
| 8     | DITHER       | Enable ordered dither.                               |
| 9     | RGB_WRITE    | Enable RGB writes.                                   |
| 10    | DEPTH_WRITE  | Enable depth writes.                                 |
| 11    | DITHER_2X2   | Use 2x2 dither instead of 4x4.                       |
| 12    | ALPHA_WRITE  | Enable alpha buffer writes.                          |
| 14    | DRAW_FRONT   | Draw to front buffer.                                |
| 15    | DRAW_BACK    | Draw to back buffer.                                 |
| 16    | DEPTH_SOURCE | Select depth source.                                 |
| 17    | Y_ORIGIN     | Use bottom-left Y origin when set.                   |
| 18    | ALPHA_PLANES | Preserve interpolated alpha in the target.           |
| 19    | ALPHA_DITHER | Dither alpha.                                        |
| 20-31 | DEPTH_OFFSET | Depth bias.                                          |

The default FBZ\_MODE enables RGB, depth test, depth write, and depth function LESS. If neither draw-target bit is set, the normal back draw buffer is used.

Depth and alpha function numbers:

| Value | Function |
|-------|----------|
| 0     | NEVER    |

| Value | Function     |
|-------|--------------|
| 1     | LESS         |
| 2     | EQUAL        |
| 3     | LESSEQUAL    |
| 4     | GREATER      |
| 5     | NOTEQUAL     |
| 6     | GREATEREQUAL |
| 7     | ALWAYS       |

ALPHA\_MODE bits:

| Bit   | Field     | Meaning                         |
|-------|-----------|---------------------------------|
| 0     | TEST_EN   | Enable alpha test.              |
| 1-3   | TEST_FUNC | Alpha test function.            |
| 4     | BLEND_EN  | Enable alpha blending.          |
| 5     | ANTIALIAS | Antialias flag.                 |
| 8-11  | SRC_RGB   | Source RGB blend factor.        |
| 12-15 | DST_RGB   | Destination RGB blend factor.   |
| 16-19 | SRC_A     | Source alpha blend factor.      |
| 20-23 | DST_A     | Destination alpha blend factor. |
| 24-31 | REF       | Alpha-test reference value.     |

Blend-factor numbers:

| Value | Factor                      |
|-------|-----------------------------|
| 0     | zero                        |
| 1     | source alpha                |
| 2     | colour average              |
| 3     | destination alpha           |
| 4     | one                         |
| 5     | one minus source alpha      |
| 6     | one minus colour average    |
| 7     | one minus destination alpha |
| 15    | source-alpha saturate       |

## 9.6 Fog, chroma, stipple, and constants

| Address            | Name           | Purpose                               |
|--------------------|----------------|---------------------------------------|
| \$F8140 to \$F823C | FOG_TABLE_BASE | 64 fog-table entries.                 |
| \$F81C4            | FOG_COLOR      | Fog RGB.                              |
| \$F81C8            | ZA_COLOR       | Z/A constant.                         |
| \$F81CC            | CHROMA_KEY     | Chroma-key colour.                    |
| \$F81D0            | CHROMA_RANGE   | Chroma-key range.                     |
| \$F81D4            | STIPPLE        | Stipple pattern.                      |
| \$F81D8            | COLOR0         | Constant colour 0, used by fast fill. |
| \$F81DC            | COLOR1         | Constant colour 1.                    |

V00D00 FOG COLOR  $r, g, b$  and V00D00 CHROMAKEY COLOR  $r, g, b$  pack the triplet as  $(b \ll 16) \mid (g \ll 8) \mid r$ . Fog blends the triangle colour towards the fog colour by the fragment Z value clamped to 0.0 to 1.0. Chroma key discards a pixel if the final RGB colour matches the key within a one-count tolerance; if CHROMA\_RANGE is non-zero, the key becomes the low RGB bound and range becomes the high RGB bound.

Stipple uses the low 32 bits as a 4 row by 8 column mask. A zero stipple word lets every pixel through.

## 9.7 Texture unit

| Address            | Name                   | Purpose                               |
|--------------------|------------------------|---------------------------------------|
| \$F8300            | TEXTURE_MODE           | Texture mode and format.              |
| \$F8304            | TL0D                   | Level-of-detail control.              |
| \$F8308            | TDETAIL                | Detail texture control.               |
| \$F830C to \$F832C | TEX_BASE0 to TEX_BASE8 | Base address per level.               |
| \$F8330            | TEX_WIDTH              | Upload width.                         |
| \$F8334            | TEX_HEIGHT             | Upload height.                        |
| \$F8338            | TEX_UPLOAD             | Commit texture memory to the sampler. |
| \$F8348            | TEX_SRC_PTR            | Guest RAM bulk texture source.        |
| \$F834C            | TEX_SRC_BYTES          | Bulk texture source byte count.       |
| \$F8400 to \$F87FF | PALETTE_BASE           | 256 texture palette entries.          |

TEXTURE\_MODE bits:

| Bit | Field   | Meaning                     |
|-----|---------|-----------------------------|
| 0   | ENABLE  | Enable texture sampling.    |
| 1-3 | MINIFY  | Minification filter.        |
| 4   | MAGNIFY | Magnification filter.       |
| 5   | CLAMP_S | Clamp S to 0.0 through 1.0. |



| Bit  | Field       | Meaning                                |
|------|-------------|----------------------------------------|
| 6    | CLAMP_T     | Clamp T to 0.0 through 1.0.            |
| 8-11 | FORMAT      | Texture format.                        |
| 12   | CHROMA      | Texture chroma key flag.               |
| 13   | TRILINEAR   | Trilinear flag.                        |
| 14   | PERSPECTIVE | Perspective-correct S/T interpolation. |
| 15   | DETAIL      | Detail texture flag.                   |
| 16   | SEQUENCE    | Sequence flag.                         |

Texture format numbers:

| Code | Format        |
|------|---------------|
| 0    | 8-bit palette |
| 1    | YIQ           |
| 2    | A8            |
| 3    | I8            |
| 4    | AI44          |
| 5    | P8            |
| 6    | ARGB8332      |
| 7    | AI88          |
| 8    | ARGB1555      |
| 9    | ARGB4444      |
| 10   | ARGB8888      |

Upload sequence:

1. Write texels into \$D0000.
2. Set TEXTURE\_DIM w,h.
3. Set TEXTURE\_MODE.
4. Use TEXTURE\_UPLOAD.
5. Enable texturing with TEXTURE\_ON.
6. Select a texture combine path with V00D00\_COMBINE.

Upload is ignored if width or height is zero, or if  $w * h * 4$  exceeds 64 KB.

TEXTURE\_UPLOAD makes the current texture available for later triangle submissions. A triangle samples the texture that was current when TRIANGLE\_CMD queued it. Uploading a different texture afterwards does not change triangles already in the queue.

## 9.8 Colour pipeline

FBZCOLOR\_PATH selects the local colour, the other colour, and the combine function used when texturing is active.

| Bits | Field          | Meaning                                 |
|------|----------------|-----------------------------------------|
| 0-1  | RGB source     | 0 iterated, 1 texture, 2 COLOR1, 3 LFB. |
| 2-3  | Alpha source   | Same encoding as RGB.                   |
| 4-6  | Combine mode   | One of the combine functions below.     |
| 27   | Texture enable | Include texture colour in the path.     |

| Combine code | Function | Meaning                    |
|--------------|----------|----------------------------|
| 0            | ZERO     | Output black.              |
| 1            | CSUB_CL  | Other minus local.         |
| 2            | ALOCAL   | Local times local alpha.   |
| 3            | AOTHER   | Local times other alpha.   |
| 4            | CLOCAL   | Local pass through.        |
| 5            | ALOCAL_T | Local alpha times texture. |
| 6            | CLOC_MUL | Local times other.         |
| 7            | AOTHER_T | Other alpha times texture. |

Convenience values accepted by V00D00 COMBINE:

| Value | Meaning                      |
|-------|------------------------------|
| 0     | Iterated vertex colour.      |
| 1     | Texture colour.              |
| 97    | Texture times vertex colour. |

## 9.9 Examples

### 9.9.1 Gouraud colour fan

COLOR\_SELECT lets a program write a different colour at each vertex. This produces a smooth red, green, and blue triangle.

```

10 REM V00D00 GOURAUD FAN
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008200
50 V00D00 CLEAR &HFF000018
60 VERTEX A 320,60
70 VERTEX B 560,410
80 VERTEX C 80,410
90 POKE32 &H000F8088,0
100 V00D00 TRICOLOR 4096,0,0,4096
110 POKE32 &H000F8088,1
120 V00D00 TRICOLOR 0,4096,0,4096
130 POKE32 &H000F8088,2
140 V00D00 TRICOLOR 0,0,4096,4096
150 TRIANGLE
160 V00D00 SWAP

```

Expected result: a large triangle blends smoothly between red at the top, green at the lower right, and blue at the lower left.

### 9.9.2 Depth-tested overlap

This draws a far blue triangle first and a nearer yellow triangle second. With depth function LESS, the nearer triangle wins where the two overlap.

```
10 REM V00D00 Z OVERLAP
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008630
50 V00D00 CLEAR &HFF000000
60 VERTEX A 220,80
70 VERTEX B 520,390
80 VERTEX C 80,390
90 V00D00 TRICOLOR 0,0,4096,4096
100 V00D00 TRIDEPTH 3277,0,0
110 TRIANGLE
120 VERTEX A 330,120
130 VERTEX B 560,350
140 VERTEX C 160,350
150 V00D00 TRICOLOR 4096,4096,0,4096
160 V00D00 TRIDEPTH 819,0,0
170 TRIANGLE
180 V00D00 SWAP
```

Line 40 sets DRAW\_BACK, DEPTH\_WRITE, RGB\_WRITE, DEPTH\_ENABLE, and depth function LESS. 3277 is about 0.8 in 20.12 fixed point. 819 is about 0.2.

### 9.9.3 Texture upload and textured triangle

This builds a four-pixel texture in texture memory and maps it across a triangle.

```

10 REM V00D00 2 BY 2 TEXTURE
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00008200
50 V00D00 CLEAR &HFF080808
60 POKE32 &H000D0000,&HFF0000FF
70 POKE32 &H000D0004,&HFF00FF00
80 POKE32 &H000D0008,&HFFFF0000
90 POKE32 &H000D000C,&HFFFFFFF
100 TEXTURE DIM 2,2
110 TEXTURE MODE &H0A61
120 TEXTURE UPLOAD
130 TEXTURE ON
140 V00D00 COMBINE 1
150 VERTEX A 320,60
160 VERTEX B 570,410
170 VERTEX C 70,410
180 POKE32 &H000F8088,0
190 V00D00 TRICOLOR 4096,4096,4096,4096
200 V00D00 TRIUV 0,0,0,0,0,0
210 POKE32 &H000F8088,1
220 V00D00 TRICOLOR 4096,4096,4096,4096
230 V00D00 TRIUV 262144,0,0,0,0,0
240 POKE32 &H000F8088,2
250 V00D00 TRICOLOR 4096,4096,4096,4096
260 V00D00 TRIUV 0,262144,0,0,0,0
270 TRIANGLE
280 V00D00 SWAP

```

&H0A61 enables texturing, selects ARGB8888 format, and clamps S and T. The texture words are written as little-endian RGBA texels. Expected result: the triangle samples red, green, blue, and white texels.

## 9.9.4 Linear aperture inspection

V00D00 PIXEL writes a 32-bit word into the texture/LFB aperture. It is useful for building texture memory and for checking address maths.

```

10 REM V00D00 PIXEL INSPECT
20 V00D00 ON
30 V00D00 DIM 320,200
40 V00D00 PIXEL 10,5,42
50 PRINT PEEK32(&H000D1928)

```

The printed value is 42, because  $(5 * 320 + 10) * 4$  is 6440, and  $\$D0000 + 6440$  is  $\$D1928$ .

## 9.9.5 Alpha, fog, chroma key, and dither

This draws translucent fogged colour over a green keyed triangle. The green triangle is discarded by the chroma key; the magenta triangle is drawn with alpha blending and dither.

```

10 REM V00D00 PIPELINE FLAGS
20 V00D00 ON
30 V00D00 DIM 640,480
40 POKE32 &H000F8110,&H00048202
50 V00D00 CLEAR &HFF202020
60 V00D00 CHROMAKEY COLOR 0,255,0
70 V00D00 CHROMAKEY ON
80 V00D00 DITHER ON
90 V00D00 FOG COLOR 40,40,80
100 V00D00 FOG ON
110 POKE32 &H000F810C,&H00005110
120 VERTEX A 320,80
130 VERTEX B 540,390
140 VERTEX C 100,390
150 V00D00 TRICOLOR 0,4096,0,4096
160 V00D00 TRIDEPTH 3000,0,0
170 TRIANGLE
180 V00D00 CHROMAKEY OFF
190 VERTEX A 330,110
200 VERTEX B 520,360
210 VERTEX C 140,360
220 V00D00 TRICOLOR 4096,0,4096,2048
230 V00D00 TRIDEPTH 1800,0,0
240 TRIANGLE
250 V00D00 SWAP

```

Line 40 keeps alpha planes so the 2048 alpha on line 220 reaches the blend unit. Line 110 sets blend enable, source factor SRC\_ALPHA, and destination factor ONE\_MINUS\_SRC\_ALPHA. Expected result: the green triangle leaves the cleared background untouched, while the magenta triangle appears softened by fog, alpha, and dither.

## 9.10 Side effects and limits

Voodoo has these programming boundaries:

| Action                | Side effect or limit                                                                                                                                                                                                               |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| V00D00 OFF            | Voodoo contributes no picture to the compositor.                                                                                                                                                                                   |
| V00D00 DIM w, h       | Ignored unless both dimensions are positive and no larger than 800 by 600.                                                                                                                                                         |
| V00D00 CLEAR colour   | Clears the drawing buffer and resets depth to a far value for LESS style depth functions.                                                                                                                                          |
| TRIANGLE              | Queues one triangle and latches its raster state and current texture. A full 4096-triangle batch is rendered into the drawing buffer without publishing, then the new triangle starts the next batch.                              |
| V00D00 SWAP           | Hands queued triangles to the rasteriser, sets busy while render or publish work runs, swaps buffers, publishes the frame, and clears the batch. Up to two swap jobs may be active; a later swap waits when that pipeline is full. |
| SWAP_BUFFER_CMD bit 1 | Clears the drawing buffer after the swap using current COLOR0.                                                                                                                                                                     |
| POKE8 to registers    | Updates the shadow byte immediately, but command side effects run only when byte 3 of the word is written.                                                                                                                         |
| Texture upload        | Copies $w * h * 4$ bytes from \$D00000 if the size fits in 64 KB.                                                                                                                                                                  |
| Texture sampling      | Uses point sampling, with wrap by default and clamp when CLAMP_S or CLAMP_T is set.                                                                                                                                                |

| Action       | Side effect or limit                                                                                |
|--------------|-----------------------------------------------------------------------------------------------------|
| Chroma key   | Discards a final pixel colour that matches the key or keyed range.                                  |
| Fog          | Blends by the clamped Z value.                                                                      |
| Dither       | Quantises RGB through an ordered matrix before writing.                                             |
| Draw targets | Front, back, or both may be selected; if no target bit is set, the normal back draw buffer is used. |

Degenerate triangles with zero area are ignored. The rasteriser clips to the picture bounds and then to the scissor rectangle when clipping is enabled. Back-facing triangles are reordered internally so that clockwise and anticlockwise vertex order both draw.